

Assignment – 9

1. Create a new Firebase project in the Firebase Console, then initialize it in your React app by installing the firebase npm package and adding firebaseConfig to a separate firebase.js file.
2. Store your Firebase API keys and config values in a .env file using REACT_APP_ prefix, and update your firebase.js to load these values from process.env.

Hint: Use process.env.REACT_APP_FIREBASE_API_KEY and similar for each config value.
3. Connect your React app to Firebase by initializing the Firebase app in firebase.js and exporting the firebase instance, then import and log the firebase object in App.js to confirm successful integration.
4. Write a short explanation (3-4 lines) of how using Firebase as a backend service benefits front-end developers building apps like Instagram or Zomato.
5. Use the Firebase documentation or ChatGPT to find out how to add Firebase Authentication to your React app, and write down the steps you would follow (no code needed, just the step-by-step process).

Answer: -

Step 1: Create a Firebase Project

Go to Firebase Console

Click Create a Project.

Enter a project name and complete the setup.

Click Add App and choose the Web (</>) option.

Register your app and copy the Firebase configuration object.

Step 2: Install Firebase in React

npm install firebase

Step 3: Create firebase.js

```
// src/firebase.js
```

```
import { initializeApp } from "firebase/app";
```

```
const firebaseConfig = {
  apiKey: "AIzaSyDuzrblSgnCQ_nOHJUKcYtPNz-1k5rsobo",
  authDomain: "react-firebase-3a941.firebaseio.com",
  projectId: "react-firebase-3a941",
  storageBucket: "react-firebase-3a941.firebaseiostorage.app",
  messagingSenderId: "744439850234",
  appId: "1:744439850234:web:8a54a8b2c01532b11d543",
  measurementId: "G-QYESDFS1GQ"
};
```

```
const app = initializeApp(firebaseConfig);
```

```
export default app;
```

2. Store Firebase Configuration in .env

Create a .env file in the project root:

```
REACT_APP_FIREBASE_API_KEY= AIzaSyDuzrblSgnCQ_nOHJUKcYtPNz-1k5rsobo
REACT_APP_FIREBASE_AUTH_DOMAIN= react-firebase-3a941.firebaseio.com
REACT_APP_FIREBASE_PROJECT_ID= react-firebase-3a941
REACT_APP_FIREBASE_STORAGE_BUCKET= react-firebase-3a941.firebaseiostorage.app
REACT_APP_FIREBASE_MESSAGING_SENDER_ID=744439850234
REACT_APP_FIREBASE_APP_ID=1:744439850234:web:8a54a8b2c01532b11d543
```

Update firebase.js:

```
import { initializeApp } from "firebase/app";
```

```
const firebaseConfig = {
  apiKey: process.env. AIzaSyDuzrblSgnCQ_nOHJUKcYtPNz-1k5rsobo,
  authDomain: process.env. react-firebase-3a941.firebaseio.com,
  projectId: process.env. react-firebase-3a941,
  storageBucket: process.env. react-firebase-3a941.firebaseiostorage.app,
  messagingSenderId: process.env. 744439850234,
  appId: process.env. 1:744439850234:web:8a54a8b2c01532b111d543
};

const app = initializeApp(firebaseConfig);

export default app;
```

3. Connect React App to Firebase

firebase.js

```
import { initializeApp } from "firebase/app";

const firebaseConfig = {
  apiKey: process.env. AIzaSyDuzrblSgnCQ_nOHJUKcYtPNz-1k5rsobo,
  authDomain: process.env. react-firebase-3a941.firebaseio.com,
  projectId: process.env. react-firebase-3a941,
  storageBucket: process.env. react-firebase-3a941.firebaseiostorage.app,
  messagingSenderId: process.env. 744439850234,
  appId: process.env. 1:744439850234:web:8a54a8b2c01532b111d543
};

const app = initializeApp(firebaseConfig);

export default app;

App.js
```

```
import app from "./firebase";

function App() {
  console.log(app);

  return (
    <div>
      <h1>Firebase Connected Successfully</h1>
    </div>
  );
}

export default App;
```

If the Firebase object appears in the browser console, the integration is successful.

4. Benefits of Firebase for Front-End Developers

Firebase provides ready-to-use backend services such as authentication, databases, cloud storage, and hosting. This allows front-end developers to build full-stack applications without creating and maintaining their own server. Applications like Instagram or Zomato can use Firebase for user login, real-time data updates, image storage, and deployment, significantly reducing development time and complexity.

5. Steps to Add Firebase Authentication in React (No Code)

1. Create a Firebase project and connect it to your React application.
2. Enable Authentication from the Firebase Console.
3. Select the sign-in methods you want to support (Email/Password, Google, GitHub, etc.).
4. Install the Firebase package in your React project.
5. Initialize Firebase in a separate `firebase.js` file.
6. Import Firebase Authentication services into your application.
7. Create signup and login forms in React.
8. Use Firebase Authentication methods to register new users.

9. Implement user login functionality.
10. Add logout functionality for authenticated users.
11. Manage authentication state using React state or context.
12. Protect private routes so only logged-in users can access them.
13. Display user information after successful login.
14. Test authentication flows and handle errors such as invalid credentials or duplicate accounts.